An explicit quartic in 342 variables

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Abstract

We specify an integer polynomial of degree four in 342 total variables whose rational input projection is the integers, under the arithmetic premises of the elliptic source construction. Its complete coefficient table has 4938 nonzero monomials. Three height detectors use sixteen local tests expressed by polynomials whose summands have different valuation classes. Existing formal-logarithm coordinates and nine additional field squares supply the required powers. The construction retains the effective abscissa-sum height bound and the explicit fixed parameter 80989^{54929}. The polynomial has content one and a largest coefficient with 1791283 binary digits.

1 Statement and coefficient specification

The rational input projection of a polynomial $P \in \mathbb{Z}[X, \boldsymbol{Y}]$ is the set of $x \in \mathbb{Q}$ for which $P(x, \boldsymbol{y}) = 0$ has a solution in rational values of $\boldsymbol{Y}$. Variable counts include the input X, and degree always means total degree. The coefficient height $H(P)$ is the largest absolute integer coefficient.

For an integer $N \geq 1$ put $A_N = 80989^{2N+1}$. The exact tables `quadratic-family342.json` and `polynomial-family342.json` specify 314 primitive integer quadratic rows $Q_{N,j}$ in 342 coordinates and

$$P_N = \sum_{j=1}^{314} Q_{N,j}^2.$$

The powers of A_N are recorded separately in the tables solely to abbreviate integer coefficients. The parameter is not a variable.

Theorem 1.1. *Under the arithmetic premises in Section 2, for every integer $N \geq 27464$ and every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{341},\ P_N(x, \boldsymbol{y}) = 0.$$

Every such polynomial has degree exactly four, content one, and 4938 nonzero coordinate monomials. Its coefficient height satisfies

$$H(P_N) = 80989^{4N+2} + 1.$$

In particular, substituting $A = 80989^{54929}$ specifies a single polynomial P_{27464} with the asserted projection. Its largest coefficient has exactly 1791283 binary digits. Algebraically, the support count holds for every $N \geq 1$, and the height formula already holds for every $N \geq 23$.

2 Arithmetic premises

Put

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \quad E : y^2 = x^3 - d^2 x,$$

$$R_0 = (1+\rho, 2\rho), \qquad P = 2R_0 = \left(\frac{57-28\rho}{8}, \frac{-660+417\rho}{32}\right).$$

We retain the following arithmetic contract, stated with its exact source locations in [5, Section 2].

1. The group and index-height assertions are

$$2E(K) = \mathbb{Z}P, \qquad h(x([n]P)) \geq n^2 - 6 \quad (n \neq 0).$$

   Here h is absolute logarithmic Weil height, and σ denotes conjugation on $K/\mathbb{Q}$.

2. For $t = u/v$ in lowest terms, $v > 0$, outside $\{0, 1, 2401, -343/32\}$, let $M_5(t)$ be the product of primes $p > 76176$ with a positive odd valuation on at least one of

$$v, \quad u, \quad u - v, \quad u - 2401v, \quad 32u + 343v.$$

   The contact-height input is

$$h(t) \leq \mathcal{H} M_5(t)^{19}, \qquad \mathcal{H} = 2^{110} 210^{1800} \left(\prod_{p \leq 76176} p\right)^{19} + 128 < 2^{2093448}.$$

3. The source pole equations exclude negative odd valuations of their nonzero input b at primes outside $\{2, 3, 17\}$. They admit the specified specialization when $0 < b < 17$, its reduced denominator is a square, b is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there are at least $3, 1, 2$, respectively. Their exact normalized seven-coordinate form is retained in the coefficient table.

4. The simultaneous prime-pattern and conjugate reduction-order input holds with arbitrary fixed odd avoidance modulus, the prescribed real approximation, and the finite congruences simultaneously. The selected primes are 1 modulo $8 \cdot 3 \cdot 17$, with $v_2(d_-) = 6$, $v_2(d_+) = 2$ and $\gcd(d_-(p), d_+(p')) = 4$. Its quantitative weighted form is retained: for fixed distinct coefficients c_j, admissible patterns $p_j = u + c_j w^2$, $N'/2 \leq u \leq N'$, $1 \leq w \leq \tau(N')\sqrt{N'}$, have total weight at least $c_0 (N')^{3/2} \tau(N')$, individual weights $O((\log N')^k)$, and $\tau(N')$ may decrease arbitrarily slowly after the fixed avoidance and congruence conditions are imposed.

These are arithmetic premises, not conclusions of coefficient verification. The weighted selection and separation used here follow from the cited analytic estimates as proved in [2]; no additional prime-distribution conjecture is introduced by the local weight replacement. The effective sum-height theorem used below is proved in [6]; it is not an extra conjectural premise.

3 The local tests and the quadratic system

The four root charts are

$$x = h^2, \qquad j^2 = x^2 - d^2,$$

and give nonzero integer multiples Q_i of P. The squareclass argument establishing this assertion, including its converse and all torsion exclusions, is proved in [4, Section 2]. Write their labels as n, a, t, b and set

$$B = \max(|n|, |a|, |t|, |b|), \qquad S = \sum_{i=1}^{4} x(Q_i) = S_0 + S_1 \rho.$$

The three positive rational square scales are

$$1, \qquad S_0^2, \qquad (S_0 + S_1)^2.$$

They are nonzero because $S_0 > 12$ and $S_0 > \sqrt{2}|S_1|$.

For each detector the contact input is $t_* = a_*\lambda$ for one of these scales. Put $\delta = t_* - 1$,

$$F_*(t_*) = t_*(t_* + 1/2)(t_* + 1)(t_* - 1/2399)(t_* + 32/407),$$

$$R_* = \frac{F_*(t_*)}{a_*^5\delta^6}, \qquad q = R_*^{-1}\delta^{-4}, \qquad q_m = q^{2m-1}.$$

These rational formulas are used only when their denominators are nonzero; the actual system consists of the polynomial rows specified in the table. The exceptional value $t_* = 1$ has empty contact radical and is explicitly covered by the local proof in [3].

Write $f(Q) = 1/x(Q)$, extended by zero at the identity. The auxiliary points satisfy $U_0 = H$, $V_0 = H - P$, and $Q_i = U_i - V_i$ as group differences on E. The exact addition rows and identity-inclusive charts of [4] enforce these relations.

Define $A_i = f(U_i)$, $B_i = \sigma f(V_i)$, and

$$C_i = L_i - L_i' \quad (1 \le i \le 4), \qquad I = L_1 - \beta,$$
$$D_H = f(H), \qquad G = L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1.$$

The alternate ratios L_i' already include conjugation. The nine tests A_i, B_i have order one, C_1, I have order ten, and D_H, C_2, C_3, C_4, G have order forty. For an ordered list of tests, use the factor

$$\mathcal{T}(D_0, \dots, D_{s-1}) = 1 + \sum_{j=0}^{s-1} q_{m(D_j)}^{2^j} D_j^{2^{j+1}},$$

where $m(D_j)$ is its stated order. Thus different required orders can occur within one factor.

The detector uses the product of the factors for these five lists:

$$(D_H, C_1, B_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3),$$
$$(C_4, A_3, B_4), \quad (G, A_4, A_1).$$

Each test occurs exactly once. The additional field powers are

$$C_1^4, I^4, A_2^4, A_3^4, A_4^4, \quad B_1^8, B_0^{16}, \quad A_1^4, A_1^8.$$

In particular the sixteenth power of B_0 is obtained by squaring its existing eighth power, and the eighth power of B_1 by squaring its existing fourth power. All additional field powers use quadratic rows. The full normalization formulas and exact coordinate accounting are in [3].

At a contact prime, a nonzero term in position j has valuation 2^j times an odd integer. These valuations are pairwise distinct and different from zero. The unique-minimum argument and the literal trace-clearing rows give every assigned test order at the same place; this is [1, Proposition 2.3]. The argument also proves the nonvanishing needed for the trace quotient. The rational logarithms, point charts, and scalar equations retain their stated forms.

3.1 A quadratic presentation of the local weights

For each detector, write $F_*(t_*) = \prod_{\theta \in \Theta}(t_* - \theta)$ and put $q = R_*^{-1}\delta^{-4}$. The weights used here are $q_m = q^{2m-1}$, so $q_{10} = q^{19}$ and $q_{40} = q^{79}$. The quadratic rows are

$$q_2 = q^2, \quad q_4 = q_2^2, \quad cq_4 = q, \quad q_8 = q_4^2,$$
$$cq_{19} = q_8^2, \quad q_{38} = q_{19}^2, \quad cq_{79} = q_{38}^2.$$

At every consumed contact prime, δ is a unit and $q \neq 0$. The rows then force $c = q^{-3}$ and each indicated power uniquely. No inverse coordinate other than c is needed. If $q = 0$, the retained

relations $\delta^3 q = e$ and $F_*(t_*)e = (a_*\delta)^5$ force $\delta = 0$; the contact radical is empty in that case. This covers the potentially degenerate stratum without using a rational formula outside its domain. The full local extraction proof from the literal trace equations is [1, Section 2].

At contacts these powers have exactly the required valuations: $v(q_m) = (1-2m)v_p(R_*)$. The ordered factors therefore supply the same orders $1, 10, 40$. At prepared finite primes their simultaneous smallness is established directly in [1, Section 3], by enlarging a fixed denominator before prime selection. Thus the replacement changes neither the formal logarithm comparisons nor their required depths.

3.2 Primitive normalization of the scalar contact product

Let z_j denote a coordinate by its source identifier. The actual contact product is the affine expression

$$\begin{aligned} r = & -103171881284 - 2103220824 z_{248} - 21050600 z_{487} \\ & - 7629700 z_{488} - 45771136 z_{489}. \end{aligned}$$

We use r itself as the coordinate in place of z_{489}. This is an invertible rational affine change: its coefficient of z_{489} is nonzero, and every other coordinate, including the input, is fixed. The source coordinate occurs only in its own defining row and the scalar normalizer row. After substitution and primitive normalization, the latter is exactly

$$R_s + 80989^2 r R_s - A_N r = 0.$$

Every equation remains quadratic. Reverse substitution reproduces both source rows exactly up to a nonzero rational scalar, so their rational solution sets correspond bijectively. Taking sums of squares preserves the required conjunction over $\mathbb{Q}$.

In the resulting quartic the coefficient of r^2 is $A_N^2 + 1$. Every other coefficient is either a nonzero constant or a nonzero constant multiple of A_N. Thus no coefficient vanishes for a positive parameter, and the support counts hold for every $N \geq 1$. Let C_0 be the largest absolute constant coefficient. Its exact integer value is recorded in the certificate and has 1520 binary digits. The largest absolute coefficient multiplying A_N is

$$L_0 = 2 \cdot 80989^2 = 13118436242.$$

The complete integer comparison verifies

$$80989^{94} + 1 > C_0, \qquad 80989^{47} > L_0, \qquad 80989^{90} + 1 < C_0.$$

Consequently, for both families,

$$H(P_N) = 80989^{4N+2} + 1 \qquad (N \geq 23).$$

Indeed this coefficient exceeds both C_0 and $L_0 A_N$ at $N = 23$, and continues to do so as N increases. Since $C_0 > L_0^2$, the exact height for any positive parameter is $\max(C_0, A_N^2 + 1)$. This is an algebraic height threshold; the sufficient parameter in the 342-variable defining theorem remains 27464, and the corresponding threshold for the 297-variable family remains ineffective.

4 Soundness with the explicit parameter

The input label and the scalar normalizer are

$$\beta = 51X^2 + 2504, \qquad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1),$$
$$R_s = \frac{A_N r}{1 + 80989^2 r}, \qquad c_s = R_s + 9792.$$

The retained scalar argument gives, with $c = |a-1|$,

$$c \geq 80989^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \leq \mathcal{H} c^{19/24}.$$

Its exceptional-point exclusions precede every use of the height detectors and are unaffected by their polynomial factors. The integer certificates in [4] give, for $N \geq 27464$,

$$|a| > (40\mathcal{H})^{10}, \qquad \log(48\mathcal{H}) < |a|/40, \qquad h(\beta) < |a|/40. \tag{1}$$

The effective sum-height theorem yields

$$B^2 \leq 2h(S_0) + 2h(S_0 + S_1) + 82.$$

Combining it with the contact-height input for the three scales gives: if a nonzero integer J is divisible by the kth powers of all three contact radicals, then

$$B^2 \leq 5\mathcal{H}|J|^{19/k}. \tag{2}$$

The formal logarithm comparison uses only the orders $1, 10, 40$, which the displayed factors supply unchanged. Therefore the fortieth powers of the contact radicals divide

$$\Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1.$$

If $n - \beta \neq 0$, their tenth powers divide its nonzero reduced numerator J.

Proof of soundness in Theorem 1.1. A rational zero of P_N makes all its quadratic rows zero. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (2) with $k = 40$ gives

$$B^2 \leq 5\mathcal{H}(14B^4)^{19/40} < 20\mathcal{H}B^{19/10}.$$

Thus $B^{1/10} < 20\mathcal{H}$, contradicting $B \geq |a|$ and (1). Hence $\Delta = 0$.

The integral Lucas descent in [5, Section 5] applies to all nonzero integer b and $|a| > 7$, giving $\log B > |a| - 3$. If $n - \beta \neq 0$, its reduced numerator satisfies $|J| \leq 2B\exp(h(\beta))$. Equation (2) with $k = 10$ would imply

$$\log B < 10\log(20\mathcal{H}) + 19h(\beta) < 29|a|/40,$$

contrary to the Lucas bound. Thus $n = \beta \in \mathbb{Z}$. Write $X = u/v$ in lowest terms with $v > 0$. Integrality of $51X^2 + 2504$ implies $v^2 \mid 51$. Since 51 is squarefree, $v = 1$ and $X \in \mathbb{Z}$. □

5 Completeness and exact verification

Fix an integer input and the scalar parameter. The retained scalar and Lucas construction supplies the integer labels and nonzero root points. Choose compact smaller real and finite-place neighborhoods avoiding the poles of the retained rational tests. The tests then have fixed valuation lower bounds at every prime in the finite preparation set S. All scales and neighborhoods are fixed before the contact primes.

Choose a positive integer D that clears the contact coefficients $D^2\theta/a_*$ and has sufficiently large valuations at the primes in S. In the simultaneous prime pattern put

$$\lambda = \frac{u}{D^2 w^2}.$$

The zero contact makes u a selected prime, outside S. Therefore $v_p(\lambda) \leq -2v_p(D)$. Making this upper bound sufficiently negative gives $v_p(t_*) < \min(0, v_p(\theta) : \theta \neq 0)$ and hence

$$v_p(F_*(t_*)) = 5v_p(t_*), \quad v_p(e) = 5v_p(a_*), \quad v_p(q) = 2v_p(a_*) - 3v_p(\lambda).$$

The latter is uniformly arbitrarily large. Thus all new weighted test terms are arbitrarily small at the prepared finite places. Enlarging D is a fixed dilation of the prime pattern and is allowed by the retained selection theorem; see [2].

At each selected contact prime, the designated primary place gives all test orders. The distinct-valuation argument at the opposite place makes $\mathrm{Tr}(W^{-1})$ a unit congruent to 1 or 2 modulo $p > 2$. Consequently the pole input $R_* \,\mathrm{Tr}(W^{-1})^{-2}$ has valuation one. The exact product formula

$$R_* = \frac{\prod_{\theta\in\Theta}(u - (\theta/a_*)D^2w^2)}{(D^2w^2)^5\delta^6}$$

shows that its valuation is even at every other prime. This includes all primes dividing δ; no unit assertion about δ away from contacts is required. Its reduced denominator is therefore a square. The selected-prime congruences give the required local squareclasses at $2, 3, 17$, and the fixed choice of D gives their required valuations.

Uniformly on the fixed compact real neighborhoods,

$$R_* \sim a_*^{-6}\lambda^{-1}, \qquad q \sim a_*^2\lambda^{-3}, \qquad \mathrm{Tr}(W^{-1}) \longrightarrow 2.$$

All factors tend to 1, and the pole input tends to 0^+. It therefore lies in $(0, 17)$ for sufficiently large λ. The simultaneous selection supplies such parameters after all fixed data have been chosen. The pole-realization premise gives fresh witnesses, and the rational weight identities fill every new quadratic row. Equal scales may reuse witnesses. This is the full simultaneous completeness argument of [1, Proposition 3.2]; it imposes no new lower bound on the scalar parameter N.

The exact reproduction programs are

```
python build_quartic_shared_weights.py
python check_quartic_shared_weights.py
```

The checker imports no producer. It checks every changed and retained row, all coordinate incidences, and every coefficient of the quartic by an independent unordered-pair expansion. It verifies degree four, content one and the coefficient-height bound. These finite checks do not prove the retained global arithmetic inputs.

The exact table contains 314 rows, 342 total coordinates, 4939 terms before parameter powers are collected, and 4938 ordinary monomials for every $N \geq 1$. The coordinate q_8 also supplies the factor term in position three; the same value is used in the shared weight presentation. Direct integer evaluation at $A = 80989^{54929}$ verifies that the largest coefficient has 1791283 binary digits. It is the coefficient of r^2.

References

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[4] *A quartic in 370 variables with specified integer coefficients*, 8 November 2026, complete quadratic specification and effective height proof. PDF; source: `quartic370_20261108_v10.tex`.

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[7] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.